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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 48.5

$$I = \int \ln \left(\frac{x^2 + 1}{x^2 - 1} \right) dx$$

Partiële integratie:

$$\begin{cases} u = \ln \left(\frac{x^2 + 1}{x^2 - 1} \right) \\ dv = dx \end{cases} \Rightarrow \begin{cases} du = \left(\frac{2x}{x^2 + 1} - \frac{2x}{x^2 - 1} \right) dx = \frac{-4x}{x^4 - 1} dx \\ v = x \end{cases}$$

$$I = x \ln \left(\frac{x^2 + 1}{x^2 - 1} \right) - \int x \cdot \frac{-4x}{x^4 - 1} dx$$

$$= x \ln \left(\frac{x^2 + 1}{x^2 - 1} \right) + \int \frac{4x^2}{x^4 - 1} dx$$

$$x^4 - 1 = (x - 1)(x + 1)(x^2 + 1)$$

$$\frac{4x^2}{x^4 - 1} = \frac{A}{x - 1} + \frac{B}{x + 1} + \frac{Cx + D}{x^2 + 1}$$

$$A = \frac{4x^2}{(x + 1)(x^2 + 1)} \Big|_{x=1} = \frac{4}{2 \cdot 2} = 1$$

$$B = \frac{4x^2}{(x - 1)(x^2 + 1)} \Big|_{x=-1} = \frac{4}{-2 \cdot 2} = -1$$

$$Ci + D = \frac{4x^2}{x^2 - 1} \Big|_{x=i} = \frac{-4}{-2} = 2$$

$$\Rightarrow C = 0, \quad D = 2$$

$$\frac{4x^2}{x^4 - 1} = \frac{1}{x - 1} - \frac{1}{x + 1} + \frac{2}{x^2 + 1}$$

$$\int \frac{4x^2}{x^4 - 1} dx = \int \left(\frac{1}{x - 1} - \frac{1}{x + 1} + \frac{2}{x^2 + 1} \right) dx$$

$$= \ln |x - 1| - \ln |x + 1| + 2 \operatorname{Bgtan} x + c$$

$$= \ln \left| \frac{x - 1}{x + 1} \right| + 2 \operatorname{Bgtan} x + c$$

Dus:

$$I = x \ln \left(\frac{x^2 + 1}{x^2 - 1} \right) + \ln \left| \frac{x - 1}{x + 1} \right| + 2 \operatorname{Bgtan} x + c$$

References